KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5083A | 5 Credits: 4 | Program Elective | Source-grounded

Industrial Electronics and Drives

 To give basic principles of different power semiconductor diodes, its working,

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5083A is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5083A.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Instrumentation Engineering
Course code	5083A
Course title	Industrial Electronics and Drives
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

15 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To give basic principles of different power semiconductor diodes, its working,
- characteristics and applications.
- To get a basic knowledge on the working of DC and AC motors.

Course outcomes

CO	Official outcome	Study level
CO1	15 Understanding semiconductor devices Explain the working and application of SCR in	See official syllabus
CO2	15 Understanding switching circuits	See official syllabus
CO3	Explain the working principle of DC & AC Motors 14 Understanding Explain single phase and three phase dual	See official syllabus
CO4	14 Understanding converters, choppers and electric drives.	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3 2
CO3	CO3 3
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain the working and application of power semiconductor devices.	4	As specified
Module 2	Explain the working and application of SCR in switching circuits	4	As specified
Module 3	Explain the working principle of DC & AC Motors	4	As specified
Module 4	Explain single phase and three phase dual converters	3	As specified

MODULE 1

Explain the working and application of power semiconductor devices.

This module is presented from the official syllabus scope for Industrial Electronics and Drives. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the working and application of power semiconductor devices.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

3 Understanding of power diodes

3 Understanding of power diodes is an official topic in Module 1 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding of power diodes using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding of power diodes should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding of power diodes means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] 3 Understanding of power diodes [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe the characteristics of SCR

Describe the characteristics of SCR is an official topic in Module 1 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the characteristics of SCR using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the characteristics of scr should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the characteristics of SCR means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] Describe the characteristics of SCR [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe the working of TRIAC and DIAC

Describe the working of TRIAC and DIAC is an official topic in Module 1 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the working of TRIAC and DIAC using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the working of triac and diac should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the working of TRIAC and DIAC means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Describe the working of TRIAC and DIAC □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Describe different protection methods of SCR 4 Understanding Power Semiconductor Devices - Power diode - Structure and VI characteristics - SCR - Construction and features - operation - transistor analogy - characteristics - specifications - holding current - latching current - gate current - turn on time-turn off time - TRIAC - structure, working, static characteristics - DIAC - structure, working, static characteristics - Protection methods - Snubber circuit.

Describe different protection methods of SCR 4 Understanding Power Semiconductor Devices - Power diode - Structure and VI characteristics - SCR - Construction and features - operation - transistor analogy - characteristics - specifications - holding current - latching current - gate current - turn on time-turn off time - TRIAC - structure, working, static characteristics - DIAC - structure, working, static characteristics - Protection methods - Snubber circuit. is an official topic in Module 1 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe different protection methods of SCR 4 Understanding Power Semiconductor Devices - Power diode - Structure and VI characteristics - SCR - Construction and features - operation - transistor analogy - characteristics - specifications - holding current - latching current - gate current - turn on time-turn off time - TRIAC - structure, working, static characteristics - DIAC - structure, working, static characteristics - Protection methods - Snubber circuit. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe different protection methods of scr 4 understanding power semiconductor devices - power diode - structure and vi characteristics - scr -

construction and features - operation - transistor analogy - characteristics - specifications - holding current - latching current - gate current - turn on time-turn off time - triac - structure, working, static characteristics - diac - structure, working, static characteristics - protection methods - snubber circuit. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe different protection methods of SCR 4 Understanding Power Semiconductor Devices - Power diode - Structure and VI characteristics - SCR - Construction and features - operation - transistor analogy - characteristics - specifications - holding current - latching current - gate current - turn on time-turn off time - TRIAC - structure, working, static characteristics - DIAC - structure, working, static characteristics - Protection methods - Snubber circuit. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Describe different protection methods of SCR 4 Understanding Power Semiconductor Devices - Power diode - Structure and VI characteristics - SCR - Construction and features - operation - transistor analogy - characteristics - specifications - holding current - latching current - gate current - turn on time-turn off time - TRIAC - structure, working, static characteristics - DIAC - structure, working, static characteristics - Protection methods - Snubber circuit. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the working and application of power semiconductor devices.
             Define semiconductor and describe the working
  M1.01                                                              3
Understanding
             of power diodes
  M1.02      Describe the characteristics of SCR                     4
Understanding
  M1.03      Describe the working of TRIAC and DIAC                  4
Understanding

  M1.04      Describe different protection methods of SCR            4
Understanding

Contents:
Power Semiconductor Devices - Power diode - Structure and VI characteristics - SCR
-
Construction and features - operation - transistor analogy - characteristics -
specifications -
holding current - latching current - gate current - turn on time-turn off time -
TRIAC -
structure, working, static characteristics - DIAC - structure, working, static
characteristics -
Protection methods - Snubber circuit.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Explain the working and application of SCR in switching circuits

This module is presented from the official syllabus scope for Industrial Electronics and Drives. Use the topic cards for learning and the source transcript for exact coverage.

As specified **CO-linked**

official hours | see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the working and application of SCR in switching circuits
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Describe triggering methods of SCR

Describe triggering methods of SCR is an official topic in Module 2 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe triggering methods of SCR using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe triggering methods of scr should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe triggering methods of SCR means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Describe triggering methods of SCR □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

Explain commutation techniques of SCR

Explain commutation techniques of SCR is an official topic in Module 2 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain commutation techniques of SCR using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain commutation techniques of scr should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain commutation techniques of SCR means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Explain commutation techniques of SCR [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe operation of inverters

Describe operation of inverters is an official topic in Module 2 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe operation of inverters using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe operation of inverters should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe operation of inverters means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] Describe operation of inverters [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe static switch and PWM 3 Understanding Converters - Triggering and Commutation of SCR - Triggering of SCR - different methods of turn on - R triggering - RC triggering - triggering using UJT relaxation oscillator - Commutation of SCRs - Line commutation and forced commutation -Operation of single- phase and three phase inverters-TRIAC light dimming circuit - Static switches-Pulse width modulation (PWM)- PWM Chip-SG352

Describe static switch and PWM 3 Understanding Converters - Triggering and Commutation of SCR - Triggering of SCR - different methods of turn on - R triggering - RC triggering - triggering using UJT relaxation oscillator - Commutation of SCRs - Line commutation and forced commutation -Operation of single- phase and three phase inverters-TRIAC light dimming circuit - Static switches-Pulse width modulation (PWM)- PWM Chip-SG352 is an official topic in Module 2 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe static switch and PWM 3 Understanding Converters - Triggering and Commutation of SCR - Triggering of SCR - different methods of turn on - R triggering - RC triggering - triggering using UJT relaxation oscillator - Commutation of SCRs - Line commutation and forced commutation -Operation of single- phase and three phase inverters-TRIAC light dimming circuit - Static switches-Pulse width modulation (PWM)- PWM Chip-SG352 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe static switch and pwm 3 understanding converters - triggering and commutation of scr - triggering of scr - different methods of turn on - r triggering - rc triggering - triggering using ujt relaxation oscillator - commutation of scrs

- line commutation and forced commutation -operation of single- phase and three phase inverters-triac light dimming circuit - static switches-pulse width modulation (pwm)- pwm chip-sg352 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe static switch and PWM 3 Understanding Converters - Triggering and Commutation of SCR - Triggering of SCR - different methods of turn on - R triggering - RC triggering - triggering using UJT relaxation oscillator - Commutation of SCRs - Line commutation and forced commutation -Operation of single- phase and three phase inverters-TRIAC light dimming circuit - Static switches-Pulse width modulation (PWM)- PWM Chip-SG352 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Describe static switch and PWM 3 Understanding Converters - Triggering and Commutation of SCR - Triggering of SCR - different methods of turn on - R triggering - RC triggering - triggering using UJT relaxation oscillator - Commutation of SCRs - Line commutation and forced commutation -Operation of single- phase and three phase inverters-TRIAC light dimming circuit - Static switches-Pulse width modulation (PWM)- PWM Chip-SG352 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the working and application of SCR in switching circuits
  M2.01      Describe triggering methods of SCR                      4
Understanding
  M2.02      Explain commutation techniques of SCR                   4
Understanding
  M2.03      Describe operation of inverters                         4
Understanding
  M2.04      Describe static switch and PWM                          3
Understanding
             Series Test - I                                         1

Contents:
Converters - Triggering and Commutation of SCR - Triggering of SCR - different
methods
of turn on - R triggering - RC triggering - triggering using UJT relaxation
oscillator -
Commutation of SCRs - Line commutation and forced commutation -Operation of single-
phase and three phase inverters-TRIAC light dimming circuit - Static switches-Pulse
width
modulation (PWM)- PWM Chip-SG352
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain the working principle of DC & AC Motors

This module is presented from the official syllabus scope for Industrial Electronics and Drives. Use the topic cards for learning and the source transcript for exact coverage.

As specified	CO-linked
official hours	see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the working principle of DC & AC Motors
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Describe the working of DC motors

Describe the working of DC motors is an official topic in Module 3 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the working of DC motors using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the working of dc motors should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the working of DC motors means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] Describe the working of DC motors [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe the classification of DC motors

Describe the classification of DC motors is an official topic in Module 3 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the classification of DC motors using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the classification of dc motors should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the classification of DC motors means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Describe the classification of DC motors □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

Describe the working of Induction motors 4 Understanding Explain the working of servo motors and

Describe the working of Induction motors 4 Understanding Explain the working of servo motors and is an official topic in Module 3 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the working of Induction motors 4 Understanding Explain the working of servo motors and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the working of induction motors 4 understanding explain the working of servo motors and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the working of Induction motors 4 Understanding Explain the working of servo motors and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ Describe the working of Induction motors 4 Understanding Explain the working of servo motors and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding stepper motors DC & AC Motors: Principle of operations of DC motor - Fleming’s left hand rule - back emf - torque equation - load characteristics of shunt and series motors - single phase induction motor - Three phase induction motor-Universal motor - application of single phase Induction motor - methods of speed control of induction motors - variable voltage - variable frequency - stepper motors - principle - applications - servo motors - tachogenerators - principle-operation - application - Soft start of AC and DC motors

4 Understanding stepper motors DC & AC Motors: Principle of operations of DC motor - Fleming’s left hand rule - back emf - torque equation - load characteristics of shunt and series motors - single phase induction motor - Three phase induction motor-Universal motor - application of single phase Induction motor - methods of speed control of induction motors - variable voltage - variable frequency - stepper motors - principle - applications - servo motors - tachogenerators - principle-operation - application - Soft start of AC and DC motors is an official topic in Module 3 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding stepper motors DC & AC Motors: Principle of operations of DC motor - Fleming’s left hand rule - back emf - torque equation - load characteristics of shunt and series motors - single phase induction motor - Three phase induction motor-Universal motor - application of single phase Induction motor - methods of speed control of induction motors - variable voltage - variable frequency - stepper motors - principle - applications - servo motors - tachogenerators - principle-operation - application - Soft start of AC and DC motors using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding stepper motors dc & ac motors: principle of operations of dc motor - fleming’s left hand rule - back emf - torque equation - load characteristics of shunt and series motors - single phase induction motor - three phase induction motor-universal motor - application of single phase induction motor - methods of speed control of induction motors - variable voltage - variable frequency - stepper motors - principle - applications - servo motors - tachogenerators - principle-operation - application - soft start of ac and dc motors should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding stepper motors DC & AC Motors: Principle of operations of DC motor - Fleming’s left hand rule - back emf - torque equation - load characteristics of shunt and series motors - single phase induction motor - Three phase induction motor-Universal motor - application of single phase Induction motor - methods of speed control of induction motors - variable voltage - variable frequency - stepper motors - principle - applications - servo motors - tachogenerators - principle-operation - application - Soft start of AC and DC motors means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 4 Understanding stepper motors DC & AC Motors: Principle of operations of DC motor - Fleming’s left hand rule - back emf - torque equation - load characteristics of shunt and series motors - single phase induction motor - Three phase induction motor-Universal motor - application of single phase Induction motor - methods of speed control of induction motors - variable voltage - variable frequency - stepper motors - principle - applications - servo motors - tachogenerators - principle-operation - application - Soft start of AC and DC motors □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the working principle of DC & AC Motors
  M3.01     Describe the working of DC motors                   2
Understanding
  M3.02     Describe the classification of DC motors            4
Understanding
  M3.03     Describe the working of Induction motors            4
Understanding
            Explain the working of servo motors and
  M3.04                                                         4
Understanding
            stepper motors

Contents:
DC & AC Motors: Principle of operations of DC motor - Fleming's left hand rule - back
emf - torque equation - load characteristics of shunt and series motors - single phase
induction motor - Three phase induction motor-Universal motor - application of single
phase Induction motor - methods of speed control of induction motors - variable voltage -
variable frequency - stepper motors - principle - applications - servo motors -
tachogenerators - principle-operation - application - Soft start of AC and DC
motors
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Explain single phase and three phase dual converters

This module is presented from the official syllabus scope for Industrial Electronics and Drives. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain single phase and three phase dual converters
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain the working of converters.

Explain the working of converters. is an official topic in Module 4 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the working of converters. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the working of converters. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the working of converters. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Explain the working of converters. [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Describe the working of choppers

Describe the working of choppers is an official topic in Module 4 of Industrial Electronics and Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Describe the working of choppers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, describe the working of choppers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Describe the working of choppers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-[illegible] Describe the working of choppers [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

Compare AC and DC Drives 4 Understanding Control Drives: Dual converters - Single phase and three phase dual converters - principle and operation waveforms - Application - Cycloconverter for single phase - applications. Choppers - Methods of output control - principle of step up and step down chopper - Jone’s chopper principle and working - waveforms. AC chopper - application of choppers - Definition of drives - comparison between AC and DC drive - Requirement of variable speed drive - Speed control of DC drives. Text / Reference: T/R Book Title/Author R1 B.L.Theraja, A Text Book of Electrical Eng volume 2, S Chand Publishers R2 P. S. Bimbhra, Power Electronics, Khanna Publishers R3 Biswanath Paul, Industrial Electronics and Control ,PHI Publishers R4 Bimal.K.Bose, Modern Power Electronics and AC Drives, PHI Publishers Joseph Vithayathil, Power Electronics Principles and Applications, McGraw Hill R5 Publishers S.K.Bhatacharya, S.chaterjee , Industrial Electronics and Control, Technical R6 Education Series Online Resources: Sl. No Website Link 1 https://www.tutorialspoint.com/power_electronics/index.htm 2 https://www.electronics-tutorials.ws/power/thyristor.html 3 https://nptel.ac.in/courses/108108077/ 4 https://www.elprocus.com/electric-drive-types-block-diagram-classification/ 5 https://automationforum.in/t/what-are-electrical-drives/4841

Compare AC and DC Drives 4 Understanding Control Drives: Dual converters - Single phase and three phase dual converters - principle and operation waveforms - Application - Cycloconverter for single phase - applications. Choppers - Methods of output control - principle of step up and step down chopper - Jone’s chopper principle and working - waveforms. AC chopper - application of choppers - Definition of drives - comparison between AC and DC drive - Requirement of variable speed drive - Speed control of DC drives. Text / Reference: T/R Book Title/Author R1 B.L.Theraja, A Text Book of Electrical Eng volume 2, S Chand Publishers R2 P. S. Bimbhra, Power Electronics, Khanna Publishers R3 Biswanath Paul, Industrial Electronics and Control ,PHI Publishers R4 Bimal.K.Bose, Modern Power Electronics and AC Drives, PHI Publishers Joseph Vithayathil, Power Electronics Principles and Applications, McGraw Hill R5 Publishers S.K.Bhatacharya, S.chaterjee , Industrial Electronics and Control, Technical R6 Education Series Online Resources: Sl. No Website Link 1 https://www.tutorialspoint.com/power_electronics/index.htm 2 https://www.electronics-tutorials.ws/power/thyristor.html 3 https://nptel.ac.in/courses/108108077/ 4 https://www.elprocus.com/electric-drive-types-block-diagram-classification/ 5 https://automationforum.in/t/what-are-electrical-drives/4841 is an official topic in Module 4 of Industrial Electronics and

Drives. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Compare AC and DC Drives 4 Understanding Control Drives: Dual converters - Single phase and three phase dual converters - principle and operation waveforms - Application - Cycloconverter for single phase - applications. Choppers - Methods of output control - principle of step up and step down chopper - Jone's chopper principle and working - waveforms. AC chopper - application of choppers - Definition of drives - comparison between AC and DC drive - Requirement of variable speed drive - Speed control of DC drives. Text / Reference: T/R Book Title/Author R1 B.L.Theraja, A Text Book of Electrical Eng volume 2, S Chand Publishers R2 P. S. Bimbhra, Power Electronics, Khanna Publishers R3 Biswanath Paul, Industrial Electronics and Control ,PHI Publishers R4 Bimal.K.Bose, Modern Power Electronics and AC Drives, PHI Publishers Joseph Vithayathil, Power Electronics Principles and Applications, McGraw Hill R5 Publishers S.K.Bhatacharya, S.chaterjee , Industrial Electronics and Control, Technical R6 Education Series Online Resources: Sl. No Website Link 1 https://www.tutorialspoint.com/power_electronics/index.htm 2 https://www.electronics-tutorials.ws/power/thyristor.html 3 https://nptel.ac.in/courses/108108077/ 4 https://www.elprocus.com/electric-drive-types-block-diagram-classification/ 5 https://automationforum.in/t/what-are-electrical-drives/4841 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, compare ac and dc drives 4 understanding control drives: dual converters - single phase and three phase dual converters - principle and operation waveforms - application - cycloconverter for single phase - applications. choppers - methods of output control - principle of step up and step down chopper - jone's chopper principle and working - waveforms. ac chopper - application of choppers - definition of drives - comparison between ac and dc drive - requirement of variable speed drive - speed control of dc drives. text / reference: t/r book title/author r1 b.l.theraja, a text book of electrical eng volume 2, s chand publishers r2 p. s. bimbhra, power electronics, khanna publishers r3 biswanath paul, industrial electronics and control ,phi publishers r4 bimal.k.bose, modern power electronics and ac drives, phi publishers joseph vithayathil, power electronics principles and applications, mcgraw hill r5 publishers s.k.bhatacharya, s.chaterjee , industrial electronics and control, technical r6 education series online resources: sl. no website link 1 https://www.tutorialspoint.com/power_electronics/index.htm 2 https://www.electronics-tutorials.ws/power/thyristor.html 3 https://nptel.ac.in/courses/108108077/ 4 https://www.elprocus.com/electric-drive-types-block-diagram-classification/ 5 https://automationforum.in/t/what-are-electrical-drives/4841 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Compare AC and DC Drives 4 Understanding Control Drives: Dual converters - Single phase and three phase dual converters - principle and operation waveforms - Application - Cycloconverter for single phase - applications. Choppers - Methods of output control - principle of step up and step down chopper - Jone’s chopper principle and working - waveforms. AC chopper - application of choppers - Definition of drives - comparison between AC and DC drive - Requirement of variable speed drive - Speed control of DC drives. Text / Reference: T/R Book Title/Author R1 B.L.Theraja, A Text Book of Electrical Eng volume 2, S Chand Publishers R2 P. S. Bimbhra, Power Electronics, Khanna Publishers R3 Biswanath Paul, Industrial Electronics and Control ,PHI Publishers R4 Bimal.K.Bose, Modern Power Electronics and AC Drives, PHI Publishers Joseph Vithayathil, Power Electronics Principles and Applications, McGraw Hill R5 Publishers S.K.Bhatacharya, S.chaterjee , Industrial Electronics and Control, Technical R6 Education Series Online Resources: Sl. No Website Link 1 https://www.tutorialspoint.com/power_electronics/index.htm 2 https://www.electronics-tutorials.ws/power/thyristor.html 3 https://nptel.ac.in/courses/108108077/ 4 https://www.elprocus.com/electric-drive-types-block-diagram-classification/ 5 https://automationforum.in/t/what-are-electrical-drives/4841 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Compare AC and DC Drives 4 Understanding Control Drives: Dual converters - Single phase and three phase dual converters - principle and operation waveforms - Application - Cycloconverter for single phase - applications. Choppers - Methods of output control - principle of step up and step down chopper - Jone’s chopper principle and working - waveforms. AC chopper -

application of choppers - Definition of drives - comparison between AC and DC drive - Requirement of variable speed drive - Speed control of DC drives. Text / Reference: T/R Book Title/Author R1 B.L.Theraja, A Text Book of Electrical Eng volume 2, S Chand Publishers R2 P. S. Bimbhra, Power Electronics, Khanna Publishers R3 Biswanath Paul, Industrial Electronics and Control ,PHI Publishers R4 Bimal.K.Bose, Modern Power Electronics and AC Drives, PHI Publishers Joseph Vithayathil, Power Electronics Principles and Applications, McGraw Hill R5 Publishers S.K.Bhatacharya, S.chaterjee , Industrial Electronics and Control, Technical R6 Education Series Online Resources: Sl. No Website Link 1 https://www.tutorialspoint.com/power_electronics/index.htm 2 https://www.electronics-tutorials.ws/power/thyristor.html 3 https://nptel.ac.in/courses/108108077/ 4 https://www.elprocus.com/electric-drive-types-block-diagram-classification/ 5 https://automationforum.in/t/what-are-electrical-drives/4841 □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain single phase and three phase dual converters

  M4.01     Explain the working of converters.                    5
Understanding
  M4.02     Describe the working of choppers                      5
Understanding
  M4.03     Compare AC and DC Drives                              4
Understanding
            Series Test - II                                      1

Contents:
Control Drives: Dual converters - Single phase and three phase dual converters -
principle
and operation waveforms - Application - Cycloconverter for single phase -
applications.
Choppers - Methods of output control - principle of step up and step down chopper
- Jone’s
chopper principle and working - waveforms. AC chopper - application of choppers -
Definition of drives - comparison between AC and DC drive - Requirement of
variable
speed drive - Speed control of DC drives.
Text / Reference:

 T/R                                      Book Title/Author

  R1      B.L.Theraja, A Text Book of Electrical Eng volume 2, S Chand Publishers
  R2      P. S. Bimbhra, Power Electronics, Khanna Publishers
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 3 Understanding of power diodes. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding of power diodes*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe the characteristics of SCR. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the characteristics of SCR*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe the working of TRIAC and DIAC. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the working of TRIAC and DIAC*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe different protection methods of SCR 4 Understanding Power Semiconductor Devices - Power diode - Structure and VI characteristics - SCR - Construction and features - operation - transistor analogy - characteristics - specifications - holding current - latching current - gate current - turn on time-turn off time - TRIAC - structure, working, static characteristics - DIAC - structure, working, static characteristics - Protection methods - Snubber circuit.. (3 marks)

Answer: Begin with the standard definition or identity of *Describe different protection methods of SCR 4 Understanding Power Semiconductor Devices - Power diode - Structure and VI characteristics - SCR - Construction and features - operation - transistor analogy - characteristics - specifications - holding current - latching current - gate current - turn on time-turn off time - TRIAC - structure, working, static characteristics - DIAC - structure, working, static characteristics - Protection methods - Snubber circuit.*.

State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain Describe triggering methods of SCR. (3 marks)

Answer: Begin with the standard definition or identity of *Describe triggering methods of SCR*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Explain commutation techniques of SCR. (3 marks)

Answer: Begin with the standard definition or identity of *Explain commutation techniques of SCR*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe operation of inverters. (3 marks)

Answer: Begin with the standard definition or identity of *Describe operation of inverters*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe static switch and PWM 3 Understanding Converters - Triggering and Commutation of SCR - Triggering of SCR - different methods of turn on - R triggering - RC triggering - triggering using UJT relaxation oscillator - Commutation of SCRs - Line commutation and forced commutation -Operation of single- phase and three phase inverters-TRIAC light dimming circuit - Static switches-Pulse width modulation (PWM)- PWM Chip-SG352. (3 marks)

Answer: Begin with the standard definition or identity of *Describe static switch and PWM 3 Understanding Converters - Triggering and Commutation of SCR - Triggering of SCR - different methods of turn on - R triggering - RC triggering - triggering using UJT relaxation oscillator - Commutation of SCRs - Line commutation and forced commutation -Operation of single- phase and three phase inverters-TRIAC light dimming circuit - Static switches-Pulse width modulation (PWM)- PWM Chip-SG352*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain Describe the working of DC motors. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the working of DC motors*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe the classification of DC motors. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the classification of DC motors*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯ definition, ▯▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯.

Define or explain Describe the working of Induction motors 4 Understanding Explain the working of servo motors and. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the working of Induction motors 4 Understanding Explain the working of servo motors and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** ▯ ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯ definition, ▯▯▯▯▯▯▯ principle/steps, ▯▯▯▯▯▯▯ application ▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯▯.

Define or explain 4 Understanding stepper motors DC & AC Motors: Principle of operations of DC motor - Fleming’s left hand rule - back emf - torque equation - load characteristics of shunt and series motors - single phase induction motor - Three phase induction motor-Universal motor - application of single phase Induction motor - methods of speed control of induction motors - variable voltage - variable frequency - stepper motors - principle - applications - servo motors - tachogenerators - principle-operation - application - Soft start of AC and DC motors. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding stepper motors DC & AC Motors: Principle of operations of DC motor - Fleming’s left hand rule - back emf - torque equation - load characteristics of shunt and series motors - single phase induction motor - Three phase induction motor-Universal motor - application of single phase Induction motor - methods of speed control of induction motors - variable voltage - variable frequency - stepper motors - principle - applications - servo motors - tachogenerators - principle-operation - application - Soft start of AC and DC motors*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Explain the working of converters.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the working of converters.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Describe the working of choppers. (3 marks)

Answer: Begin with the standard definition or identity of *Describe the working of choppers*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Compare AC and DC Drives 4 Understanding Control Drives: Dual converters - Single phase and three phase dual converters - principle and operation waveforms - Application - Cycloconverter for single phase - applications. Choppers - Methods of output control - principle of step up and step down chopper - Jone’s chopper principle and working - waveforms. AC chopper - application of choppers - Definition of drives - comparison between AC and DC drive - Requirement of variable speed drive - Speed control of DC drives. Text / Reference: T/R Book Title/Author R1 B.L.Theraja, A Text Book of Electrical Eng volume 2, S Chand Publishers R2 P. S. Bimbhra, Power Electronics, Khanna Publishers R3 Biswanath Paul, Industrial Electronics and

Control ,PHI Publishers R4 Bimal.K.Bose, Modern Power Electronics and AC Drives, PHI Publishers Joseph Vithayathil, Power Electronics Principles and Applications, McGraw Hill R5 Publishers S.K.Bhatacharya, S.chaterjee , Industrial Electronics and Control, Technical R6 Education Series Online Resources: Sl. No Website Link 1 https://www.tutorialspoint.com/power_electronics/index.htm 2 https://www.electronics-tutorials.ws/power/thyristor.html 3 https://nptel.ac.in/courses/108108077/ 4 https://www.elprocus.com/electric-drive-types-block-diagram-classification/ 5 https://automationforum.in/t/what-are-electrical-drives/4841. (3 marks)**

Answer: Begin with the standard definition or identity of *Compare AC and DC Drives 4 Understanding Control Drives: Dual converters - Single phase and three phase dual converters - principle and operation waveforms - Application - Cycloconverter for single phase - applications. Choppers - Methods of output control - principle of step up and step down chopper - Jone’s chopper principle and working - waveforms. AC chopper - application of choppers - Definition of drives - comparison between AC and DC drive - Requirement of variable speed drive - Speed control of DC drives. Text / Reference: T/R Book Title/Author R1 B.L.Theraja, A Text Book of Electrical Eng volume 2, S Chand Publishers R2 P. S. Bimbhra, Power Electronics, Khanna Publishers R3 Biswanath Paul, Industrial Electronics and Control ,PHI Publishers R4 Bimal.K.Bose, Modern Power Electronics and AC Drives, PHI Publishers Joseph Vithayathil, Power Electronics Principles and Applications, McGraw Hill R5 Publishers S.K.Bhatacharya, S.chaterjee , Industrial Electronics and Control, Technical R6 Education Series Online Resources: Sl. No Website Link 1 https://www.tutorialspoint.com/power_electronics/index.htm 2 https://www.electronics-tutorials.ws/power/thyristor.html 3 https://nptel.ac.in/courses/108108077/ 4 https://www.elprocus.com/electric-drive-types-block-diagram-classification/ 5 https://automationforum.in/t/what-are-electrical-drives/4841*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **3 Understanding of power diodes**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Describe triggering methods of SCR**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Describe the working of DC motors**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain the working of converters.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl. No Website Link
- https://www.tutorialspoint.com/power_electronics/index.htm
- https://www.electronics-tutorials.ws/power/thyristor.html
  https://nptel.ac.in/courses/108108077/
- https://www.elprocus.com/electric-drive-types-block-diagram-classification/
- https://automationforum.in/t/what-are-electrical-drives/4841

IN-PAGE SEARCH

Search this lesson